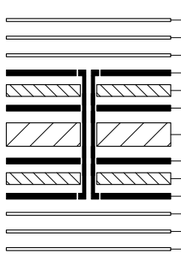
Main board Fabrication Document

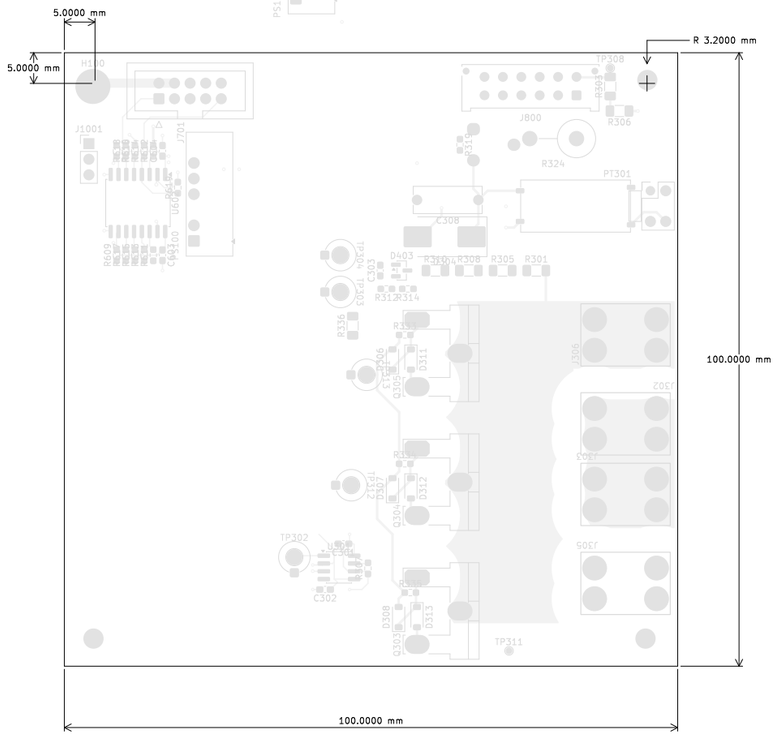
Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**F.Cu (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**In1.Cu (GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Core		1.828mm	FR4	Dielectric	
Copper	**In2.Cu (Sig, PWR)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**B.Cu (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL SNPB

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100.000 × 100.000 mm
BOARD THICKNESS	2.278 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	1.000 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.300 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm

Lim. resistor
PFC

HMI

Ext
ctrl

SWD
Status

Lamp
trig

1 kV strobe
04/Jul/2025

Boot

Cap+

Lamp+

Light
sense

Felipe L Rodrigues

Cap−

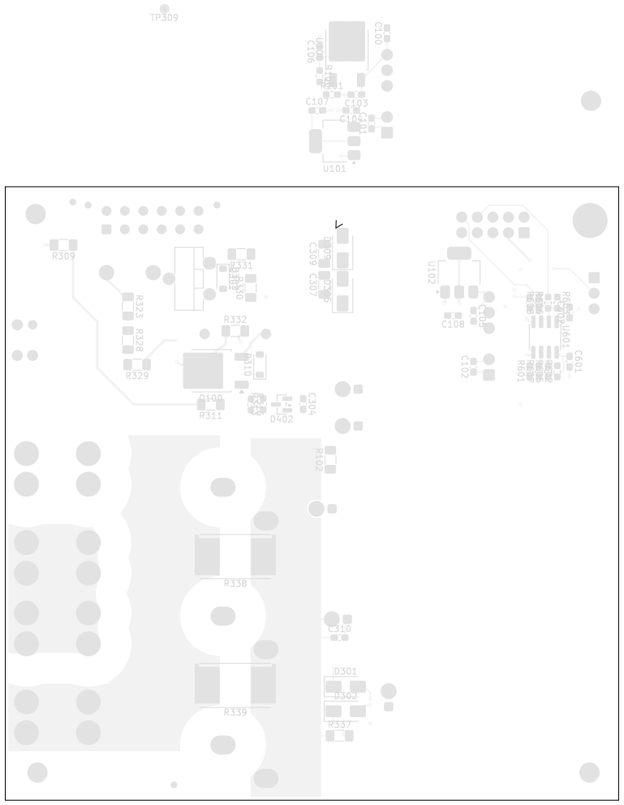
Lamp−

ChurrasCod
Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: **Main board**		Project Name: **1 kV strobe**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **1** of **10**

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



1 kV strobe
04/Jul/25
Felipe L Rodrigues

ChurrasCod Electronics

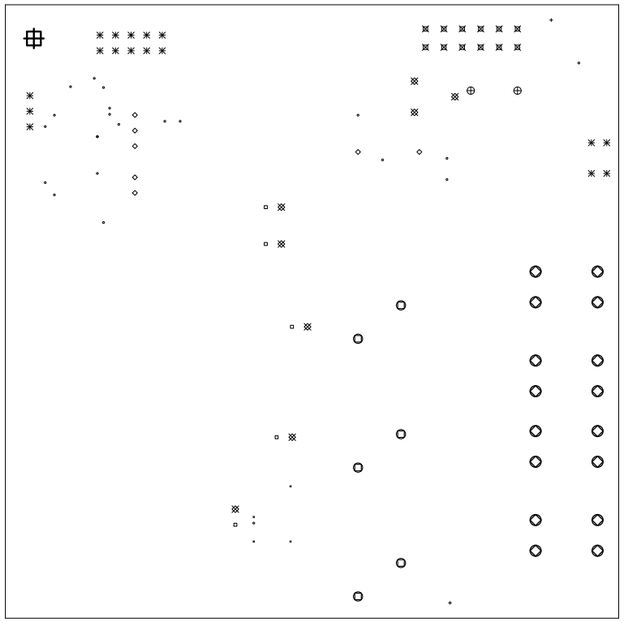
Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: **Main board**		Project Name: **1 kV strobe**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 2 of 10

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	0.25mm (9.84mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Via
○	29	0.30mm (11.81mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Via
+	24	0.50mm (19.69mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
□	5	0.70mm (27.56mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
◇	12	0.80mm (31.50mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
⊠	2	0.90mm (35.43mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
✳	20	1.00mm (39.37mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
⊠	12	1.02mm (40.16mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
⊠	8	1.10mm (43.31mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
⊕	2	1.20mm (47.24mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
○	6	1.50mm (59.06mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
○	16	1.80mm (70.87mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
⊕	1	3.20mm (125.98mils)	PTH	Round	F.Cu (Slg, PWR) - B.Cu (Slg, PWR)	Pad
	Total 141					



ChurrasCod Electronics

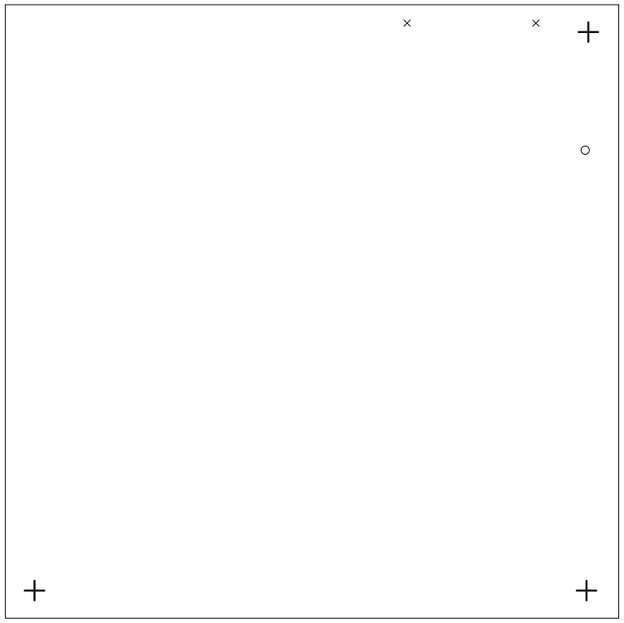
Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10

Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1.02mm (40.16mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
○	1	1.35mm (53.15mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
+	4	3.20mm (125.98mils)	NPTH	Round	F.Cu (Sig, PWR) - B.Cu (Sig, PWR)	Mechanical
	Total 7					

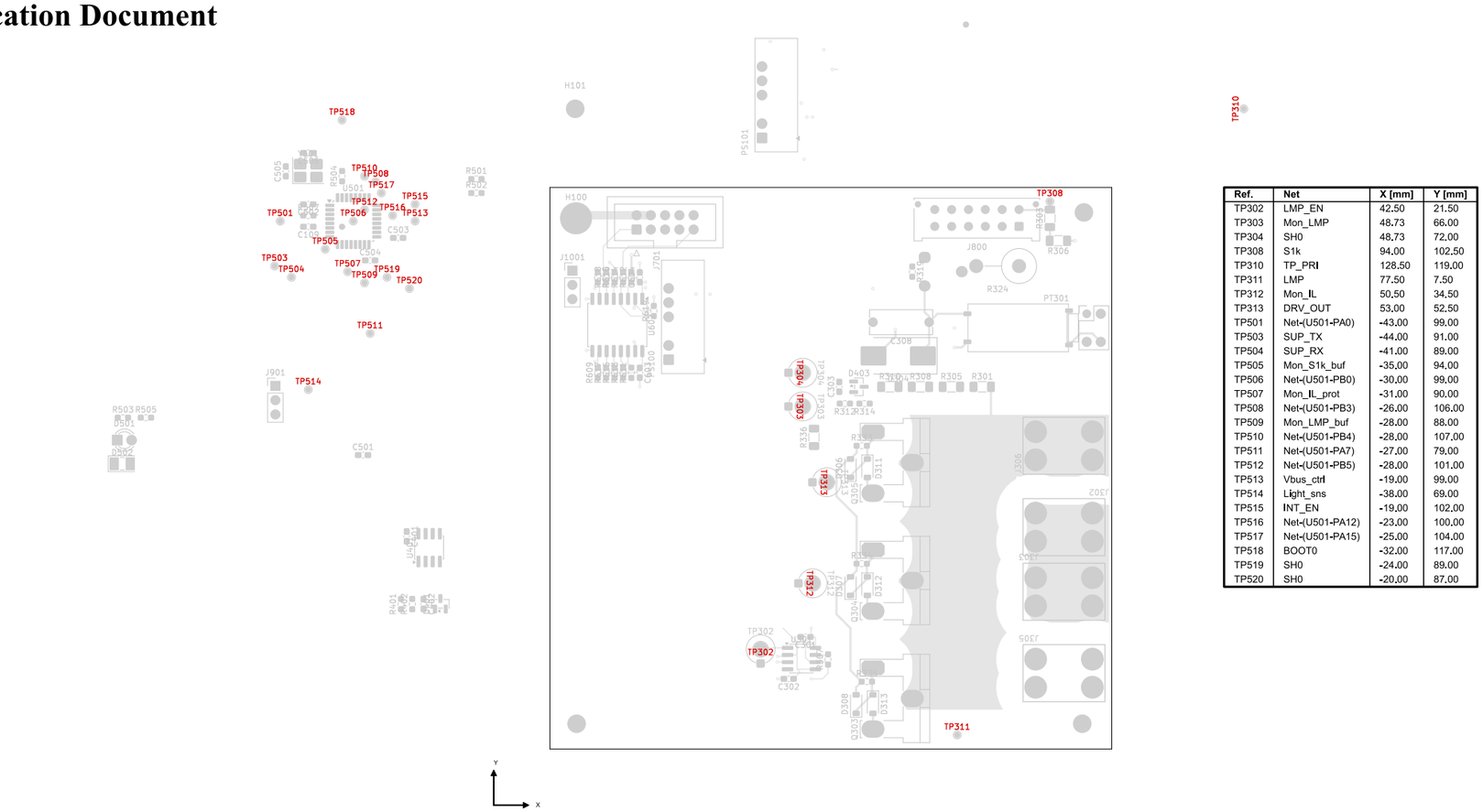


ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 4 of 10

Main board Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP302	LMP_EN	42.50	21.50
TP303	Mon_LMP	48.73	66.00
TP304	SH0	48.73	72.00
TP308	S1k	94.00	102.50
TP310	TP_PRI	128.50	119.00
TP311	LMP	77.50	7.50
TP312	Mon_IL	50.50	34.50
TP313	DRV_OUT	53.00	52.50
TP501	Net-(U501-PA0)	-43.00	99.00
TP503	SUP_TX	-44.00	91.00
TP504	SUP_RX	-41.00	89.00
TP505	Mon_S1k_buf	-35.00	94.00
TP506	Net-(U501-PB0)	-30.00	99.00
TP507	Mon_IL_prot	-31.00	90.00
TP508	Net-(U501-PB3)	-26.00	106.00
TP509	Mon_LMP_buf	-28.00	88.00
TP510	Net-(U501-PB4)	-28.00	107.00
TP511	Net-(U501-PA7)	-27.00	79.00
TP512	Net-(U501-PB5)	-28.00	101.00
TP513	Vbus_ctrl	-19.00	99.00
TP514	Light_sns	-38.00	69.00
TP515	INT_EN	-19.00	102.00
TP516	Net-(U501-PA12)	-23.00	100.00
TP517	Net-(U501-PA15)	-25.00	104.00
TP518	BOOT0	-32.00	117.00
TP519	SH0	-24.00	89.00
TP520	SH0	-20.00	87.00

Lim. resistor
PFC

HMI

Ext
ctrl

SWD
Status

Lamp
trig

1 kV strobe
04/Jul/2025

Cap+

Lamp+

Boot

Light
sense

Felipe L Rodrigues

Cap−

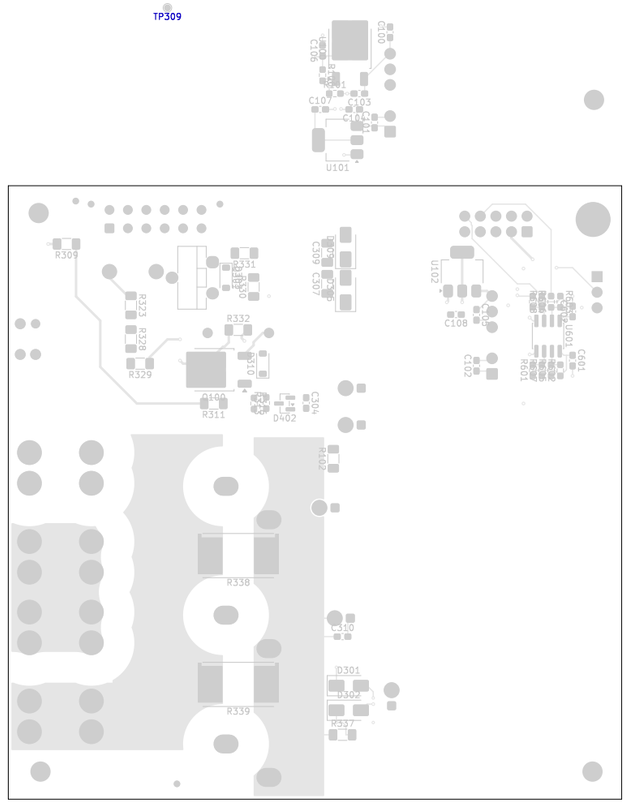
Lamp−

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Top Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 5 of 10

Main board Fabrication Document

Bottom Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP309	TP_REF	79.05	134.00
TP502	Net-(U501-PA1)	-32.00	98.00

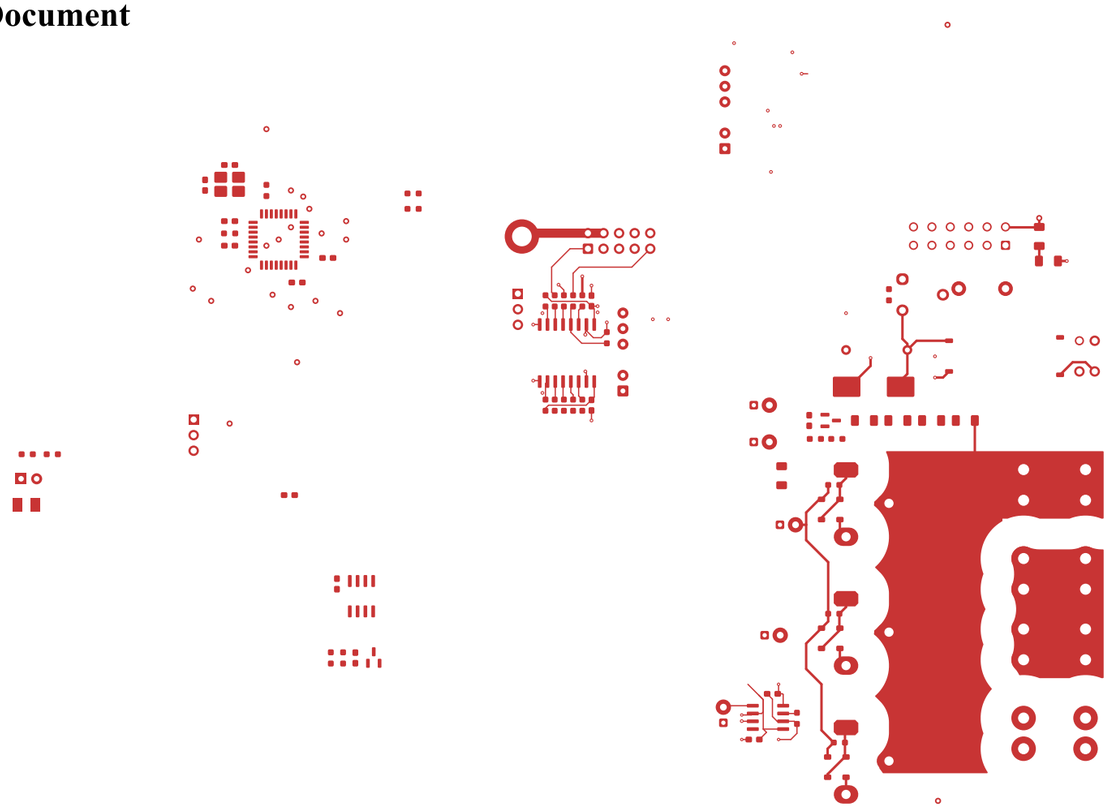
1 kV strobe
04/Jul/25
Felipe L Rodrigues

ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: **Main board**		Project Name: **1 kV strobe**	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 6 of 10

Main board Fabrication Document

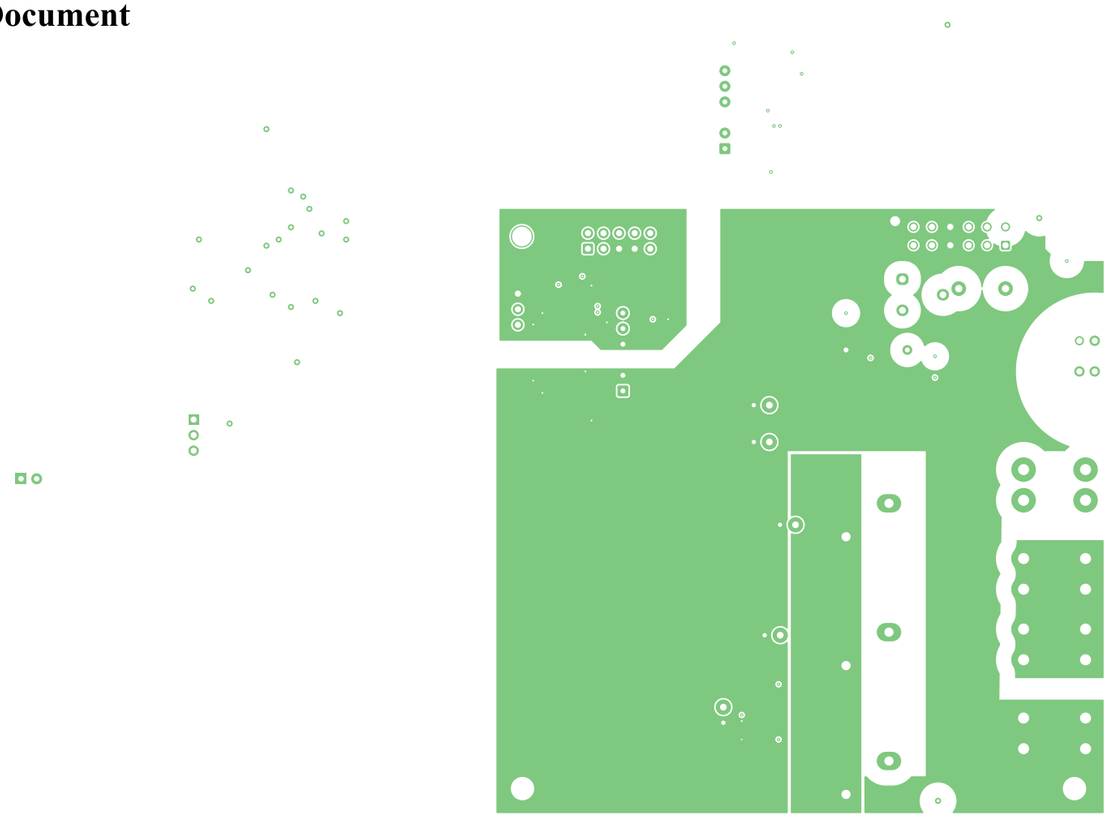
F.Cu (Sig, PWR) (Scale 1:1)



ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics	Variant: PRELIMINARY	Git Hash: 2db8fa8
		Board Name: Main board	Project Name: 1 kV strobe	
	Sheet Title: F.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb / Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:	Reviewer:	Size: A3	Sheet: 7 of 10

Main board Fabrication Document

In1.Cu (GND) (Scale 1:1)

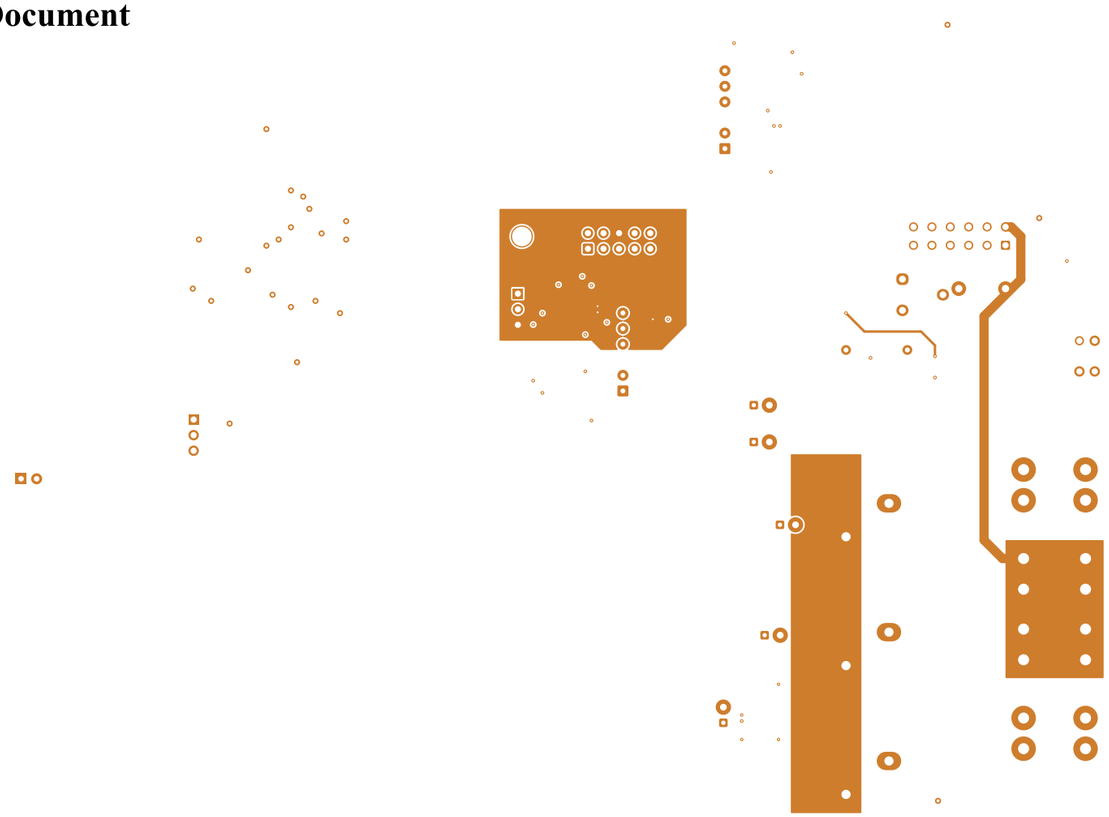


ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: In1.Cu (GND) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 8 of 10

Main board Fabrication Document

In2.Cu (Sig, PWR) (Scale 1:1)

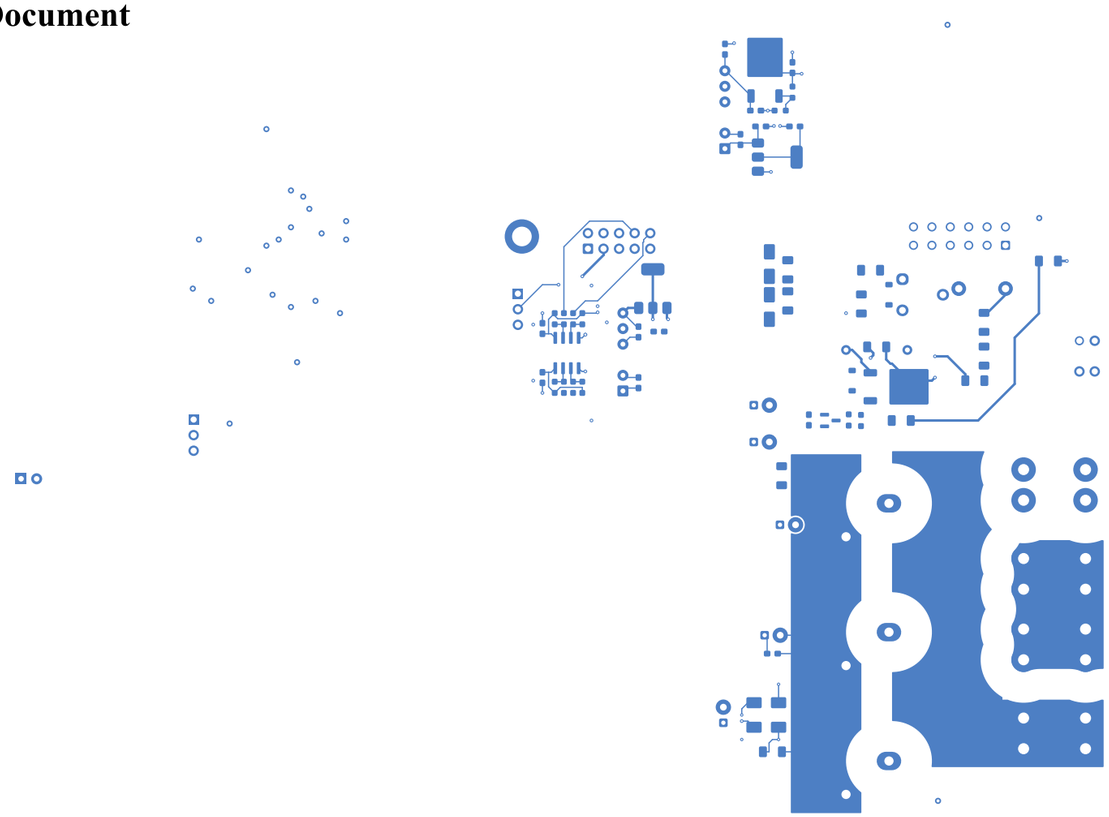


ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 9 of 10

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



ChurrasCod Electronics

Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 2db8fa8
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 10 of 10